Suppose that the domain consists of all people. Express each of these statements using quantifiers; logical connectives; and $P(x)$, $Q(x)$, $R(x)$ and $S(x)$.

a) Babies are illogical: $\forall x (P(x) \rightarrow \neg Q(x))$

b) Nobody is despised who can manage a crocodile: $\forall x (R(x) \rightarrow \neg S(x))$

c) Illogical persons are despised: $\forall x (\neg Q(x) \rightarrow S(x))$

d) Babies cannot manage crocodiles: $\forall x (P(x) \rightarrow \neg R(x))$

e) Does (d) follow from (a), (b) and (c)? If not, is there a correct conclusion?

Let, $\forall x (P(x) \rightarrow \neg Q(x))$ be True. By conditional-disjunction equivalence, $\forall x (\neg P(x) \vee \neg Q(x))$ is True. By defn. of universal quantifier, for all elements a in the domain, $\boxed{\neg P(a) \vee \neg Q(a)}$ is True.

Let $\forall x (R(x) \rightarrow \neg S(x))$ be True. By conditional-disjunction equivalence, $\forall x (\neg R(x) \vee \neg S(x))$ is True. By defn. of universal quantifier, for all elements a in the domain, $\boxed{\neg R(a) \vee \neg S(a)}$ is True.

Let $\forall x (\neg Q(x) \rightarrow S(x))$ be True. By conditional-disjunction equivalence, $\forall x (\neg(\neg Q(x)) \vee S(x))$ is True. By double-negation law, $\forall x (Q(x) \vee S(x))$ is True. By defn. of universal quantifier, for all elements a in the domain, $\boxed{Q(a) \vee S(a)}$ is True. By defn. of OR, the following cases arise: ① Both $Q(a)$ and $S(a)$ are True. By defn. of negation, $\neg Q(a)$ and $\neg S(a)$ are both False. Since $\boxed{\neg P(a) \vee \neg Q(a)}$ and $\boxed{\neg R(a) \vee \neg S(a)}$ are both True, by defn. of OR, $\neg P(a)$ is True and $\neg R(a)$ is True. By defn. of OR, $\boxed{\neg P(a) \vee \neg R(a) \text{ is True}}$. ② $Q(a)$ is True, $S(a)$ is False. By defn. of negation, $\neg Q(a)$ is False. Since $\boxed{\neg P(a) \vee \neg Q(a)}$ is True, by defn. of OR, $\neg P(a)$ is True. By defn. of OR, $\boxed{\neg P(a) \vee \neg R(a) \text{ is True}}$.

③ $Q(a)$ is False, $S(a)$ is True. By defn. of negation, $\neg S(a)$ is False. Since $\boxed{\neg R(a) \vee \neg S(a)}$ is True, by defn. of OR, $\neg R(a)$ is True.